

and new systems administrators without fear of the command line.

It abstracts away many of the deep complexities of granular sysadmin tasks behind the Red Hat Enterprise Linux Web console. RHEL 8 also includes Red Hat Enterprise Linux System Roles, which automates many of the more complex tasks around managing and configuring Linux in production. To enhance security, RHEL 8 supports the OpenSSL 1.1.1 and TLS 1.3 cryptographic standards.

RHEL 8 forms the foundation for Red Hat's entire hybrid cloud portfolio, starting with Red Hat OpenShift Container Platform 4 and the upcoming Red Hat OpenStack Platform 15. The forthcoming RHEL CoreOS, a minimal footprint OS designed to host Red Hat OpenShift Container Platform deployments, is also built on RHEL 8.

RHEL 8 is also broadly supported as a guest operating system on Red Hat hybrid cloud infrastructure, including Red Hat OpenShift 4, Red Hat OpenStack Platform 15 and Red Hat Virtualization 4.3.

Along with the launch of RHEL 8, the company also announced the general availability of the Red Hat Universal Base Image, a user space image derived from RHEL for building Red Hat certified Linux containers. The Red Hat Universal Base Image is available to all developers with or without a RHEL subscription, providing a more secure and reliable foundation for building enterprise-ready containerised applications.

TensorFlow reveals open source MLIR for rapid machine learning

MLIR, not a language like Python or C++, signifies an intermediate compilation stage between the machine code and higher-level languages. Google's engineers, who are working on the TensorFlow machine learning (ML) framework, have launched a sub-project, Multi-Level Intermediate Representation (MLIR), which is proposed to be a common intermediate language for all the ML frameworks.



The MLIR is an open source project, and a certified specification is available for those who want to create it.

MLIR will enable projects that are using TensorFlow and other libraries of ML to be collected for more efficient code. This takes the maximum advantage of original hardware. In addition, MLIR can be used

by compilers, which extends its optimisation advantages beyond the ML projects.

The LLVM compiler infrastructure project uses intermediate representation of its own. MLIR is co-created by Chris Lattner, one of LLVM's original creators. Earlier, Lattner and Googler Tatiana Shpeisman described how TensorFlow is already creating multiple intermediate representations internally. However, these disparate intermediate representations do not benefit from each other. MLIR offers a single, standard intermediate representation for all the subsystems of TensorFlow. Currently, TensorFlow is migrating to internally utilise MLIR.

In addition, MLIR can offer parallelised compilation. The sub-project is being developed to enable a compiler to work on diverse segments of code in parallel. This will allow ML models and other such applications to be more quickly pushed to production.

Microsoft announces next version of Windows Subsystem for Linux

Microsoft's newfound love for Linux and other open source projects is getting deeper. The tech giant has announced Windows Subsystem for Linux 2 (WSL 2), the next version of WSL, at its Build 2019 developers' conference.

Windows Subsystem for Linux (WSL) is a compatibility layer for running Linux binary executables natively on Windows. The upcoming version, WSL 2, is based on a Linux 4.19 kernel coming soon to Windows. This kernel uses technology built for Azure. In both cases, it helps to reduce Linux boot time and streamline memory use.

"WSL 2 also improves file system I/O performance, Linux compatibility, and can run Docker containers natively so that a VM is no longer needed for containers on Windows," said Kevin Gallo, corporate VP, Windows Developer Platform.

Windows Subsystem for Linux 2 (WSL 2) will be much faster than WSL as Microsoft promises "twice as much speed for file system-heavy operations, such as Node Package Manager install."

At the Build 2019 developers' conference, Microsoft also announced a slew of other offerings for Windows developers, including Windows Terminal, XAML Islands, React Native for Windows and MSIX Core.

Windows Terminal, available in preview now, is a new application for command line users that promises a user interface with 'graphics-processing-unit-accelerated text rendering'.

Both the Windows Terminal and WSL 2 are coming out in mid-June. In addition, Microsoft plans to launch XAML Islands, a standard Universal Windows Platform control (which was slated to have come out in May).

Meanwhile, the company has released a performance-oriented React Native for Windows implementation on GitHub. It also debuted a developer preview of MSIX Core, which enables Windows 10's MSIX packages to be installed on Windows 7 and Windows 8.



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